a statistically rare fluctuation from a normal, unsurprising, high-entropy, completely disordered configuration.

Think of it this way: if you toss a handful of pennies over and over again, sooner or later they will all land heads. If you have nearly the infinite patience needed to throw the jumbled pages of *War and Peace* in the air over and over again, sooner or later they will land in correct numerical order. If you wait with your open bottle of flat Coke, sooner or later the random jostling of the carbon dioxide molecules will cause them to reenter the bottle. And, for Boltzmann's kicker, if the universe waits long enough—for nearly an eternity, perhaps—its usual, high-entropy, highly probable, totally disordered state will, through its own bumping, jostling, and random streaming of particles and radiation, sooner or later just happen to coalesce into the configuration that we all see right now. Our bodies and brains would emerge fully formed from the chaos—stocked with memories, knowledge, and skills—even though the past they seem to reflect would never really have happened. Everything we know about, everything we value, would amount to nothing more than a rare but every-so-often-expectable statistical fluctuation momentarily interrupting a near eternity of disorder. This is schematically illustrated in Figure 6.4.

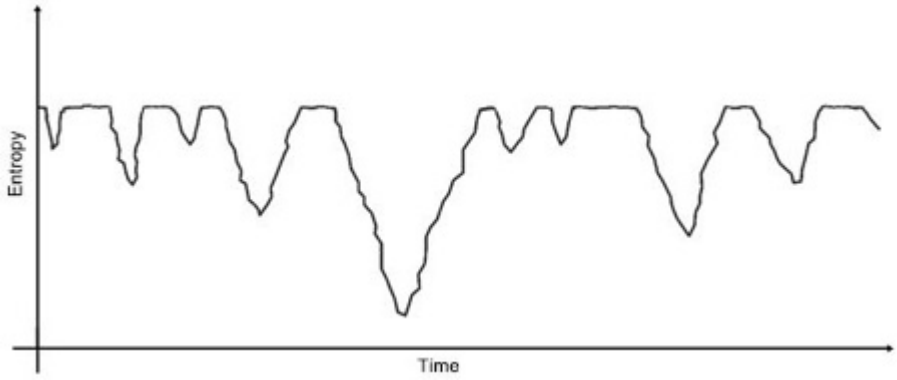


Figure 6.4 A schematic graph of the universe's total entropy through time. The graph shows the universe spending most of its time in a state of total disorder—a state of high entropy—and every so often experiencing fluctuations to states of varying degrees of order, varying states of lower entropy. The greater the entropy dip, the less likely the fluctuation. Significant dips in entropy, to the kind of order in the universe today, are extremely unlikely and would happen very rarely.